WORKED SOLUTION

STEP II 1999, Question 6

Pure • Integration by Substitution

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/99-s2-q6

The key idea

The derivative of a fraction $\frac{ax+b}{cx+d}$ is a constant over $(cx+d)^2$. So whenever an integrand is $\frac{1}{(x+3)^2}$ times a function of $\frac{x+1}{x+3}$ or $\frac{x+2}{x+3}$, substituting that fraction as the new variable absorbs the $\frac{1}{(x+3)^2}$ completely, and leaves us with nothing worse than $\int \ln y \, \mathrm{d}y$. The other two integrals then follow from the laws of logarithms.

Solution

The derivative

By the quotient rule,

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{a(cx+d) - c(ax+b)}{(cx+d)^2} = \frac{ad-bc}{(cx+d)^2}.$$

The x terms in the numerator cancel, leaving a constant. This is what makes the whole question work.

The first integral

Let $y = \dfrac{x+1}{x+3}$. This has the form $(*)$ with $a=1$, $b=1$, $c=1$, $d=3$, so

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{3-1}{(x+3)^2} = \frac{2}{(x+3)^2}, \qquad \text{that is,} \qquad \frac{\mathrm{d}x}{(x+3)^2} = \tfrac{1}{2}\,\mathrm{d}y.$$

When $x = 0$, $y = \frac{1}{3}$; when $x = 1$, $y = \frac{1}{2}$. Using $\int \ln y \, dy = y \ln y - y$ (integration by parts with $u = \ln y$):

$$\begin{aligned}\int_0^1 \frac{1}{(x+3)^2} \ln\left(\frac{x+1}{x+3}\right) dx &= \frac{1}{2}\int_{1/3}^{1/2} \ln y \, dy = \frac{1}{2}\Big[y \ln y - y\Big]_{1/3}^{1/2} \\ &= \frac{1}{2}\left(-\tfrac{1}{2}\ln 2 - \tfrac{1}{2} + \tfrac{1}{3}\ln 3 + \tfrac{1}{3}\right) \\ &= \frac{1}{6}\ln 3 - \frac{1}{4}\ln 2 - \frac{1}{12},\end{aligned}$$

using $\ln \frac{1}{2} = -\ln 2$ and $\ln \frac{1}{3} = -\ln 3$. This is the required result. Call this integral I_1.

A companion integral

Ask yourself What has $x^2 + 3x + 2$ got to do with the substitution we have just used?

It factorises as $(x+1)(x+2)$, so

$$\frac{x^2+3x+2}{(x+3)^2} = \frac{x+1}{x+3} \cdot \frac{x+2}{x+3} \qquad \text{and} \qquad \frac{x+1}{x+2} = \frac{x+1}{x+3} \Big/ \frac{x+2}{x+3}.$$

All the logarithms in the last two integrals are therefore sums and differences of $\ln \frac{x+1}{x+3}$ and $\ln \frac{x+2}{x+3}$ (all these fractions are positive for $0 \le x \le 1$). So we need one more integral,

$$J = \int_0^1 \frac{1}{(x+3)^2} \ln\left(\frac{x+2}{x+3}\right) dx.$$

Let $y = \dfrac{x+2}{x+3}$. Now $a = 1$, $b = 2$, $c = 1$, $d = 3$, so $\dfrac{dy}{dx} = \dfrac{1}{(x+3)^2}$, and the limits become $y = \frac{2}{3}$ and $y = \frac{3}{4}$:

$$\begin{aligned}J &= \int_{2/3}^{3/4} \ln y \, dy = \Big[y \ln y - y\Big]_{2/3}^{3/4} \\ &= \tfrac{3}{4}(\ln 3 - 2\ln 2) - \tfrac{3}{4} - \tfrac{2}{3}(\ln 2 - \ln 3) + \tfrac{2}{3} \\ &= \frac{17}{12}\ln 3 - \frac{13}{6}\ln 2 - \frac{1}{12}.\end{aligned}$$

The remaining two integrals

Since $\ln \frac{x^2+3x+2}{(x+3)^2} = \ln \frac{x+1}{x+3} + \ln \frac{x+2}{x+3}$, the second integral is

$$\begin{aligned}I_1 + J &= \left(\tfrac{1}{6} + \tfrac{17}{12}\right)\ln 3 - \left(\tfrac{1}{4} + \tfrac{13}{6}\right)\ln 2 - \tfrac{1}{12} - \tfrac{1}{12} \\ &= \frac{19}{12}\ln 3 - \frac{29}{12}\ln 2 - \frac{1}{6}.\end{aligned}$$

Since $\ln\frac{x+1}{x+2} = \ln\frac{x+1}{x+3} - \ln\frac{x+2}{x+3}$, the third integral is

$$\begin{aligned} I_1 - J &= \left(\tfrac{1}{6} - \tfrac{17}{12}\right)\ln 3 - \left(\tfrac{1}{4} - \tfrac{13}{6}\right)\ln 2 \\ &= \frac{23}{12}\ln 2 - \frac{5}{4}\ln 3, \end{aligned}$$

where the constant terms $-\frac{1}{12}$ have cancelled.

Checks

- Simpson's rule on $[0,1]$ gives $I_1 \approx -0.07352$, $J \approx -0.02878$, and the second and third integrals ≈ -0.10230 and -0.04473, all agreeing with the exact forms to ten decimal places.
- A rough bound: $\int_0^1 (x+3)^{-2}\,\mathrm{d}x = \frac{1}{12}$, and $\ln\frac{x+1}{x+2}$ lies between $\ln\frac{1}{2} \approx -0.69$ and $\ln\frac{2}{3} \approx -0.41$. So the third integral must lie between about -0.058 and -0.034, and -0.0447 does.
- Every integrand is negative on $[0,1]$ (each fraction inside a logarithm is less than 1), so all three answers should be negative, and they are.

Common slips

- Forgetting the factor $\frac{1}{2}$ in the first substitution: $ad - bc = 2$ there, but 1 in the second.
- Not changing the limits, or keeping x-limits 0 and 1 after substituting.
- Trying a fresh substitution for $\frac{x+1}{x+2}$: it works, but leaves $(x+3)^{-2}$ in terms of y as something messy. Splitting the logarithm is far quicker.

Beyond STEP

Maps of the form $x \mapsto \frac{ax+b}{cx+d}$ with $ad - bc \neq 0$ are *Möbius transformations*. Composing two of them corresponds to multiplying the 2×2 matrices $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$, and $ad - bc$ is the determinant, which is why it appears in the derivative. In complex analysis they are exactly the conformal bijections of the extended plane, sending circles and lines to circles and lines. Needham's *Visual Complex Analysis* has a beautiful chapter on them.

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